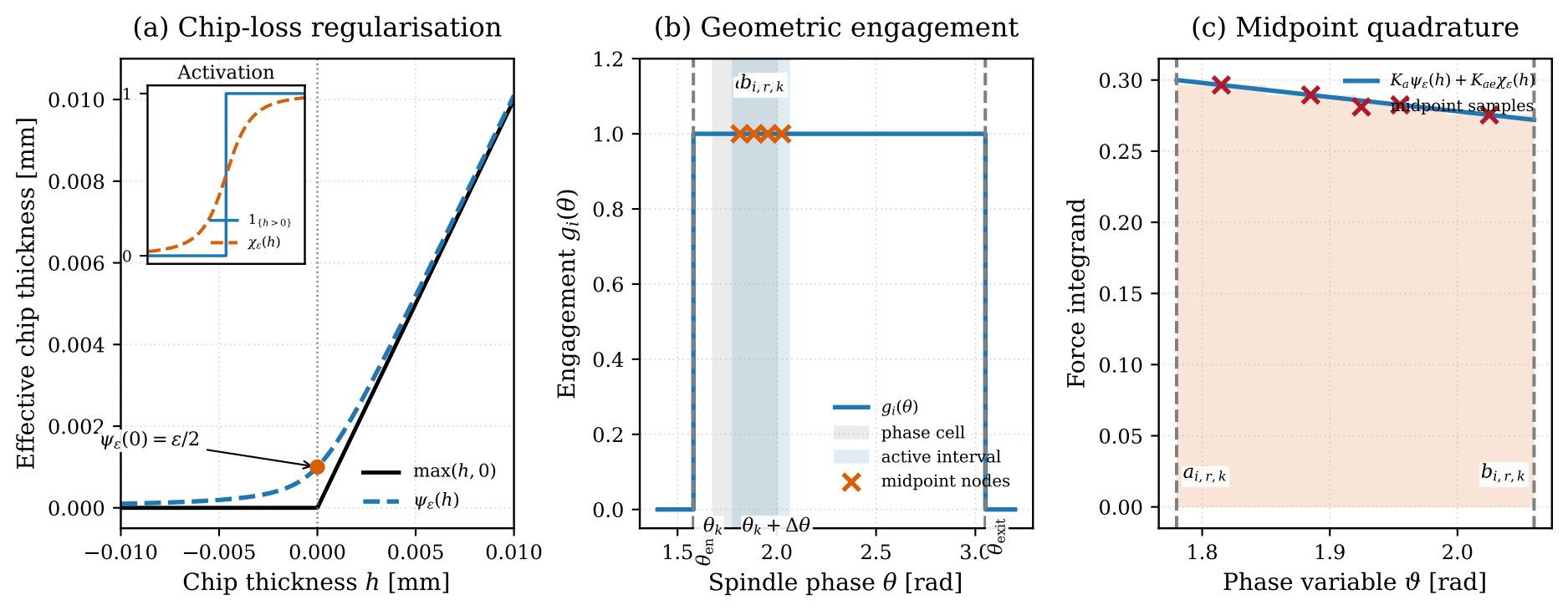
(a) Chip-loss regularisation
Activation
1
0
$1_{\{h>0\}}$
$\chi_\varepsilon(h)$
Effective chip thickness [mm]
0.010
0.008
0.006
0.004
0.002
0.000
$\psi_\varepsilon(0) = \varepsilon/2$
max(h, 0)
$\psi_\varepsilon(h)$
−0.010
−0.005
0.000
0.005
0.010
Chip thickness h [mm]
(b) Geometric engagement
$b_{i,r,k}$
Engagement $g_i(\theta)$
1.2
1.0
0.8
0.6
0.4
0.2
0.0
$g_i(\theta)$
phase cell
active interval
midpoint nodes
θ_k
$\theta_k + \Delta\theta$
θ_{en}
θ_{exit}
1.5
2.0
2.5
3.0
Spindle phase θ [rad]
(c) Midpoint quadrature
$K_a\psi_\varepsilon(h) + K_{ae}\chi_\varepsilon(h)$
midpoint samples
Force integrand
0.30
0.25
0.20
0.15
0.10
0.05
0.00
$a_{i,r,k}$
$b_{i,r,k}$
1.8
1.9
2.0
Phase variable ϑ [rad]